

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: BenQ
Firmware Version: N/A
Model(s): TH680, MH680

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.1.0

Version History

Module Version	Date	Notes
1_0_0_0	11/15/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='TH680')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='TH680')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command 3DMode	Value 'Auto' 'Frame Packing' 'Inverter Disabled'	Value 'Top and Bottom' 'Side by Side' 'Inverter Enabled'	Value 'Frame Sequential' 'Sync Off'
# 3DMode example InterfaceName.Set('3DMode', 'Auto')			
Command AspectRatio	Value 'Auto' '16:9'	Value 'Real' '16:10'	Value '4:3'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1' 'Video' 'USB Display'	Value 'Computer 2' 'S-Video' 'USB Reader'	Value 'HDMI' 'Network'
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command LampMode	Value 'Normal'	Value 'Eco 1'	Value 'Eco 2'
# LampMode example InterfaceName.Set('LampMode', 'Normal')			

Command MenuCall	Value 'On'	Value 'Off'	
# MenuCall example InterfaceName.Set('MenuCall', 'On')			
Command MenuNavigation	Value 'Up' 'Right'	Value 'Down' 'Enter'	Value 'Left'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			
Command PictureMode	Value 'Dynamic' 'Cinema'	Value 'Presentation' 'User 1'	Value 'sRGB' 'User 2'
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command VolumeStep	Value 'Up'	Value 'Down'	
# VolumeStep example InterfaceName.Set('VolumeStep', 'Up')			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command 3DMode	Value 'Auto' 'Frame Packing' 'Inverter Disabled'	Value 'Top and Bottom' 'Side by Side' 'Inverter Enabled'	Value 'Frame Sequential' 'Sync Off'
# 3DMode examples InterfaceName.Update('3DMode') Value = InterfaceName.ReadStatus('3DMode') InterfaceName.SubscribeStatus('3DMode', None, FeedbackHandler)			
Command AspectRatio	Value 'Auto' '16:9'	Value 'Real' '16:10'	Value '4:3'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute examples InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	

<pre># Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)</pre>			
Command Input	Value 'Computer 1' 'Video' 'USB Display'	Value 'Computer 2' 'S-Video' 'USB Reader'	Value 'HDMI' 'Network'
<pre># Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LampMode	Value 'Normal'	Value 'Eco 1'	Value 'Eco 2'
<pre># LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)</pre>			
Command LampUsage	Value Hours		
<pre># LampUsage examples InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)</pre>			
Command PictureMode	Value 'Dynamic' 'Cinema'	Value 'Presentation' 'User 1'	Value 'sRGB' 'User 2'
<pre># PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)</pre>			
Command Power	Value 'On'	Value 'Off'	
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command VideoMute	Value 'On'	Value 'Off'	
<pre># VideoMute examples InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)</pre>			
Command VolumeStatus	Value 0 – 10		

```
# VolumeStatus examples
InterfaceName.Update('VolumeStatus')
Value = InterfaceName.ReadStatus('VolumeStatus')
InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)
```

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 115200

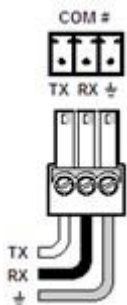
Data Bits: 8

Parity: None

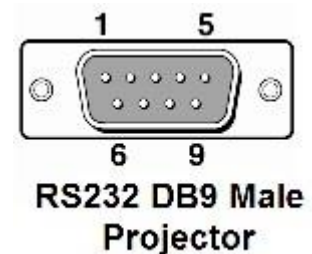
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		3	TxD
RxD		2	RxD
GND		5	GND



Appendix A. Set Commands

3D Mode Auto	\x0D*3d=auto#\x0D
3D Mode Frame Packing	\x0D*3d=fp#\x0D
3D Mode Frame Sequential	\x0D*3d=fs#\x0D
3D Mode Inverter Disabled	\x0D*3d=da#\x0D
3D Mode Inverter Enabled	\x0D*3d=iv#\x0D
3D Mode Side by Side	\x0D*3d=sbs#\x0D
3D Mode Sync Off	\x0D*3d=off#\x0D
3D Mode Top and Bottom	\x0D*3d=tb#\x0D
Aspect Ratio 16:10	\x0D*asp=16:10#\x0D
Aspect Ratio 16:9	\x0D*asp=16:9#\x0D
Aspect Ratio 4:3	\x0D*asp=4:3#\x0D
Aspect Ratio Auto	\x0D*asp=AUTO#\x0D
Aspect Ratio Real	\x0D*asp=REAL#\x0D
Audio Mute Off	\x0D*mute=off#\x0D
Audio Mute On	\x0D*mute=on#\x0D
Auto Image None	\x0D*auto#\x0D
Freeze Off	\x0D*freeze=off#\x0D
Freeze On	\x0D*freeze=on#\x0D
Input Computer 1	\x0D*sour=RGB#\x0D
Input Computer 2	\x0D*sour=RGB2#\x0D
Input HDMI	\x0D*sour=hdm#\x0D
Input Network	\x0D*sour=network#\x0D
Input S-Video	\x0D*sour=svid#\x0D
Input USB Display	\x0D*sour=usbdisplay#\x0D
Input USB Reader	\x0D*sour=usbreader#\x0D
Input Video	\x0D*sour=vid#\x0D
Lamp Mode Eco 1	\x0D*lampm=eco#\x0D
Lamp Mode Eco 2	\x0D*lampm=seco#\x0D
Lamp Mode Normal	\x0D*lampm=lnor#\x0D
Menu Call Off	\x0D*menu=off#\x0D
Menu Call On	\x0D*menu=on#\x0D
Menu Navigation Down	\x0D*down#\x0D
Menu Navigation Enter	\x0D*enter#\x0D
Menu Navigation Left	\x0D*left#\x0D
Menu Navigation Right	\x0D*right#\x0D
Menu Navigation Up	\x0D*up#\x0D

Global Scripter Module Communication Sheet

Picture Mode Cinema	\x0D*appmod=cine#\x0D
Picture Mode Dynamic	\x0D*appmod=dynamic#\x0D
Picture Mode Presentation	\x0D*appmod=preset#\x0D
Picture Mode sRGB	\x0D*appmod=srgb#\x0D
Picture Mode User 1	\x0D*appmod=user1#\x0D
Picture Mode User 2	\x0D*appmod=user2#\x0D
Power Off	\x0D*pow=off#\x0D
Power On	\x0D*pow=on#\x0D
Video Mute Off	\x0D*blank=off#\x0D
Video Mute On	\x0D*blank=on#\x0D
Volume Step Down	\x0D*vol=-#\x0D
Volume Step Up	\x0D*vol=+#\x0D

Appendix B. Update Commands

3D Mode	\x0D*3d=?#\x0D
Aspect Ratio	\x0D*asp=?#\x0D
Audio Mute	\x0D*mute=?#\x0D
Freeze	\x0D*freeze=?#\x0D
Input	\x0D*sour=?#\x0D
Lamp Mode	\x0D*lampm=?#\x0D
Lamp Usage	\x0D*ltim=?#\x0D
Picture Mode	\x0D*appmod=?#\x0D
Power	\x0D*pow=?#\x0D
Video Mute	\x0D*blank=?#\x0D
Volume Status	\x0D*vol=?#\x0D